Project 2 DATASCI 200

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GitHub repository: https://github.com/UC-Berkeley-I-School/Project2_Wu.git

Overview

The dataset we would like to investigate in this project is the political TV advertisement archive data, which includes archives political ads and tracks ad airings in select markets in the 2016 elections. These datasets provide details about airings of ads on TV, giving information about when and where they aired.

The dataset includes the following 19 columns. Description is added based on the content under each column.

columns	**description**	**data type**
id	unique numerical ID for each airing of an ad	int64
wp_identifier	unique numerical ID linked to each advertisement	int64
network	the network where the ad is distributed	object
location	location where the ad was distributed	object
program	which TV program the ad was distributed on	object
program_type	type of the TV program	object
start_time	start time of the ad	datetime64[ns, UTC]
end_time	end time of the ad	object
archive_id	unique ID linked to each advertisement	object
embed_url	an url linked to the ad being archived	object
sponsors	the sponsers for the ad	object
sponsor_types	the type of the sponsers for the ad	object
race	which race it was targeting: presidential, senate, etc.	object
cycle	which election year it was targeting	float
subjects	the subjects discussed in the ad	object
candidates	election candidate name(s)	object
type	the type of the ad	object
message	whether the ad is in support to the candidate(s) or against the candidate(s), or it conveys mixed or unknown message	object
date_created	date when the airing record is created	object

For this project, we are interested in exploring patterns and relationships regarding the variables listed below, and intend to understand and gain more insights on the TV ad

distribution strategy employed by candidates for presidential campaign purpose, through different time regime across the 2016 presidential election cycle.

1. Air time: start_time. We checked the distribution of the length of ad (end time minus start time), all ads length ranges from 14 to 300 seconds which is not very long, and thus we think it is appropriate and decide to use start_time as the air time of each ad and to decide which day the ad is located in)
2. candidates
3. message

Questions

1. Is there a trend of the number of TV ad distributed through the cycle of 2016 presidential election? Do candidates distribute more TV ad around major primary debates, presidential debates, and election day? On top of a more general analysis across all candidates, we would also like to check that for four candidates we are interested in the most (Donald Trump, Marco Rubio, Bernie Sanders, and Hilary Clinton).
2. For 2016 presidential nominees Donald Trump and Hillary Clinton, were there more TV ad in support of them, or more against them, during the 2016 presidential election cycle?

Data Cleaning/Sanity Checks

There are 375350 rows of data in the dataset, with 19 columns in total. We first conducted sanity check on the data and the following columns include nan/null data.

```
num of nan/nulls in each column
race              29872
cycle             29101
candidates        25127
subjects          22562
sponsor_types      1162
sponsors           1162
location             25
program              15
```

We then checked whether there is any duplicates in the dataset. We define duplicate ad airings as two or more airing that that has the same value in column 'start_time', 'end_time', 'network', and 'wp_identifier', but given different values in column ‘id’. No such duplicate exists in the dataset.

Based on the missing data summary and the questions we intend to answer, we checked column ‘candidates’, which we found that 1) 70899 rows of airing ads cover more than one candidate (and can be as many as eight), and 2) apart from nan values in that column representing missing candidate information, the dataset also uses ‘none’, which we will also

treat as missing candidate information in our analysis. For ads that cover more than one candidate, we looked at some of the urls provided. Most of them convey negative message from one candidate attacking other candidate(s) in 'candidate' column of that ad airing. Without looking into each advertisement or using additional data outside of this dataset, it would be hard to identify which one candidate initiate this advertisement against the other candidate(s). Based on that, we decide to drop those airings with more than one candidate.

We then conduced sanity check on column 'cycle' to make sure the data we are exploring are only for 2016 election campaign cycle, which is confirmed as there is only one value '2016.0' apart from nan in that column.

We further examined the column 'race' to make sure the data we are exploring are for presidential election only (which would be represented as 'PRES' in 'race' column), summary attached below. We see that there is around 223722 rows of ad airings that is labeled as for presidential election (including those that may contain other missing information), which is around 2/3 of the total number of data in the dataset. The size should be adequate for our analysis even after we further remove rows with missing data.

	race	id
0	AZ05	713
1	AZS1	4745
2	CAS1	437
3	CO06	5
4	COS1	2521
5	FLS2	6404
6	IAS1	1628
7	ILS2	149
8	INS2	6
9	KS01	17

	race	id
10	KYS2	7
11	MD04	1448
12	MD08	5392
13	MDS2	2202
14	NC02	208
15	NC11	8
16	NCS2	6668
17	NHS1	21680
18	NJ03	211

	race	id
19	NVS2	19451
20	OH08	830
21	OH14	336
22	OHS2	8048
23	PA02	168
24	PA08	1198
25	PAS1	26088
26	PRES	223722
27	WI01	127
28	WIS2	11061

For 'message' column, we noticed that it has both 'mix' and 'mixed' which we assume should mean the same type without further clarification, and thus we plan to convert all 'mixed' value to 'mix' in that column. 'unknown' is also one of the value showing up in that column, and we do not think it should be treated as missing value and thus still want to keep it for our analysis.

Based on the check above, we plan to do the following data cleaning and conversion:

1. Convert 'mixed' to 'mix' in column 'message'
2. Drop rows that contains nan/null/'none' values in column 'candidates', 'cycle' and 'race'
3. Drop rows that contains more than one candidate in column 'candidates'
4. Drop rows that values in 'race' is not 'PRES' (drop if not for presidential election)
5. Drop all columns except the following 3: candidates, start_time, message

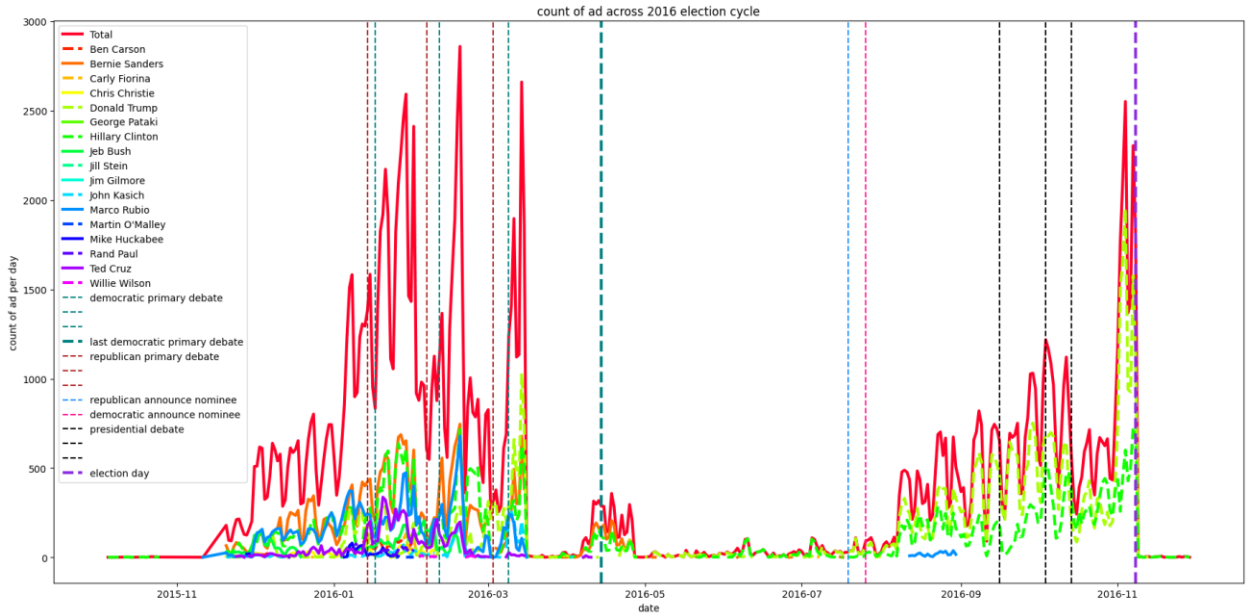
After data cleaning, the dataset contains 178353 rows and 3 columns. **Please note, we assume this is the final dataset we would like to conduct our analysis on, and will not specify "ad for 2016 presidential election" and "ad that includes only one candidate" repeatedly in the remaining sections of the report.**

Data Analysis

- Question 1:

We select the major primary debates dates as 2016-01-17, 2016-02-11, 2016-03-09, 2016-4-14 for Democratic, and 2016-01-14, 2016-02-06, 2016-03-03 for Republican. 2016 presidential debates dates are 2016-09-16, 2016-10-04, and 2016-10-14. Election day is 2016-11-08.

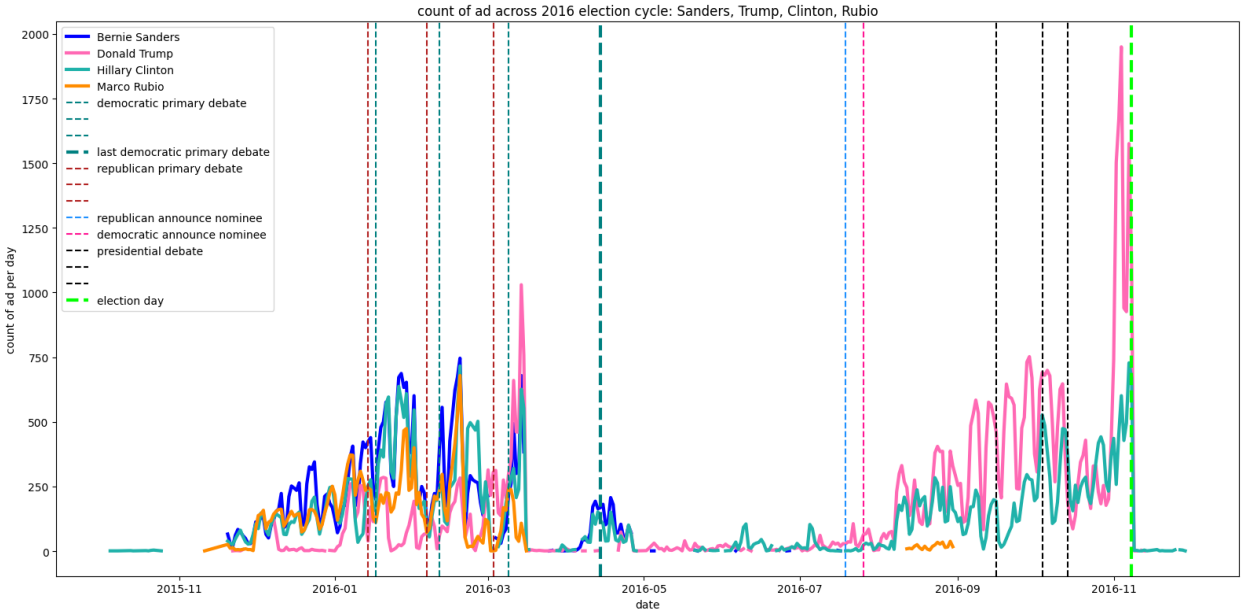
We plot below the count of ad, in total and per candidate, through the cycle of 2016 presidential election, and we can see that candidates started to distribute more ad on TV starting from mid-November 2016. It also dropped sharply after most of the major primary debates ended in both parties in mid-March 2016, until the presidential nominee was announced officially by both parties in mid-July 2016.



As we zoom in on the four candidates we are interested in the most (Donald Trump, Marco Rubio, Bernie Sanders, and Hilary Clinton), we notice that there was another increase in the number of TV ad around April 2016 mostly caused by the increase of ad distribution on three of these four candidates (Sanders, Clinton and Trump). It mostly could be because one additional Democratic debate was arranged between top Democratic candidates Bernie Sanders and Hillary Clinton on 2016/4/14 to seek for the official presidential nomination by Democratic. We also notice Marco Rubio stopped its ad distribution for presidential election, after announcing drop-out on March 15th 2016. However, there are some abnormal TV ad

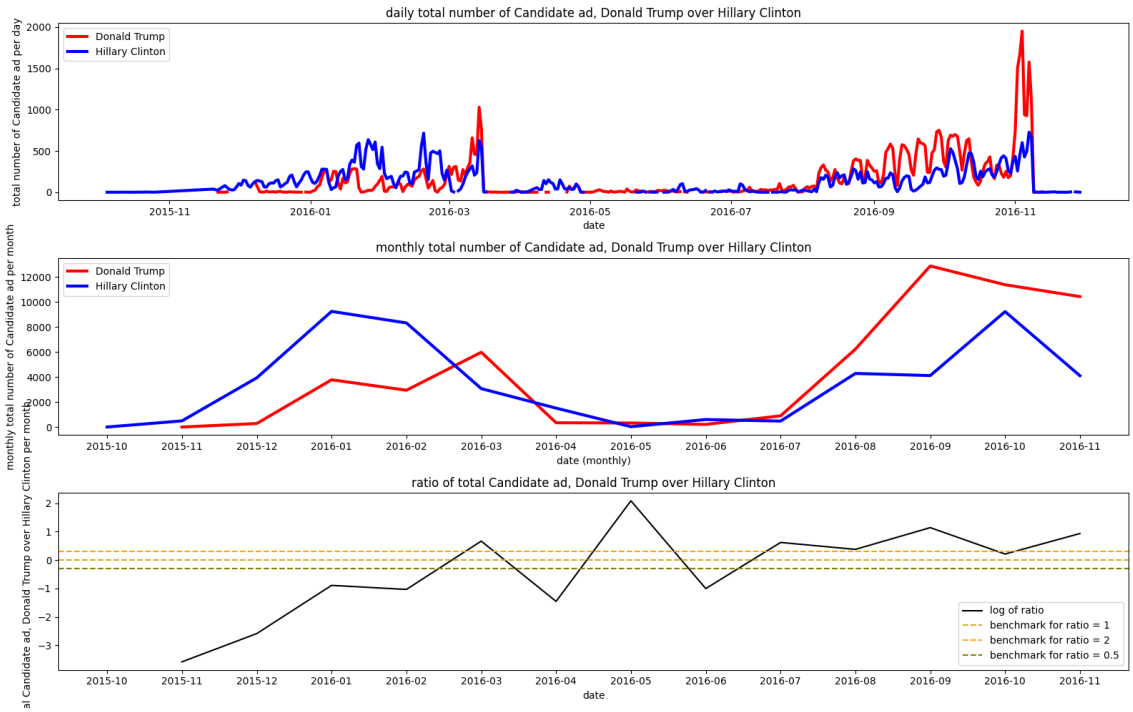
regarding him before the first presidential debate. We checked some url (for example, id 29103, https://archive.org/embed/PolAd_MarcoRubio_vz8kf) regarding to those ads and they are for Rubio running for senator, which suggests that those data may be mislabeled in 'race' column. Considering its small size, it should not affect our conclusion to Question 1, but it is still alarming and worthy of our additional attention considering it may affect our assumption on the data quality and thus potentially results in invalidity or bias on our analysis and conclusions. Especially for this case, it is almost impossible to detect this type of "wrong data", without additional information on when the candidate dropped out of the presidential election, followed by a manual check on the content in those questioned advertisements.

After the official presidential nominee was announced in July 2016, presidential nominees from both Republican and Democratic, TV ad distribution on Donald Trump and Hillary Clinton started to increase. The number of ad on those two candidates reached some local maximum right before each presidential debate, and achieved around 2000 per day before the election day.

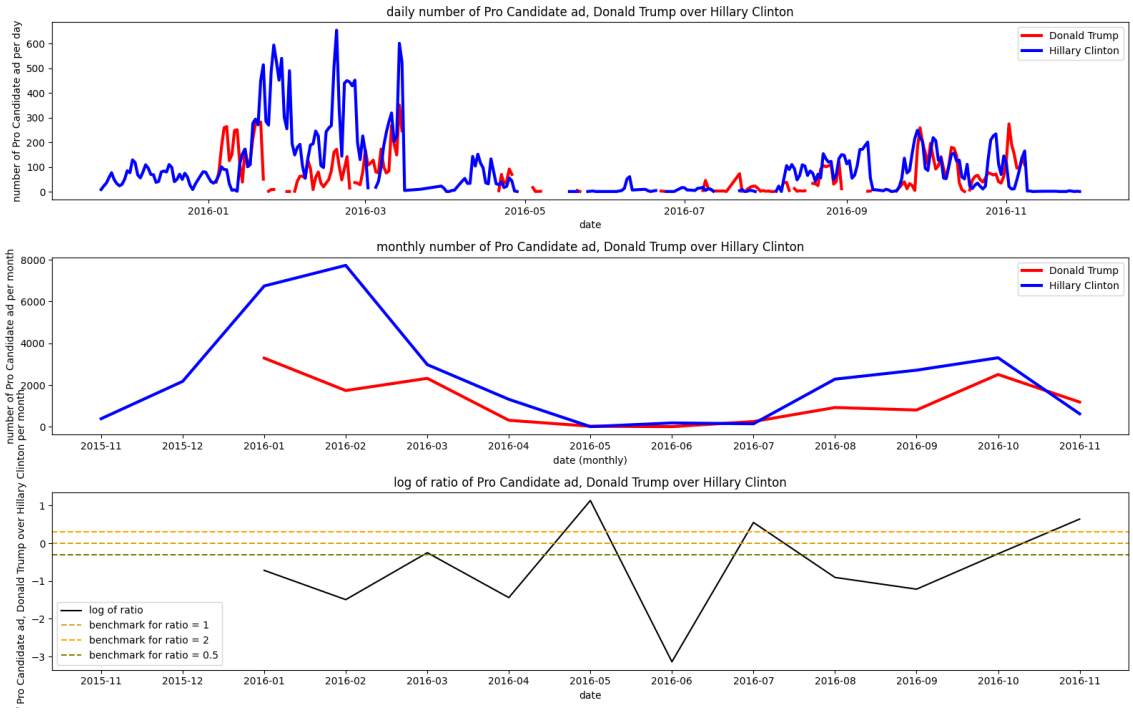


- Question 2:

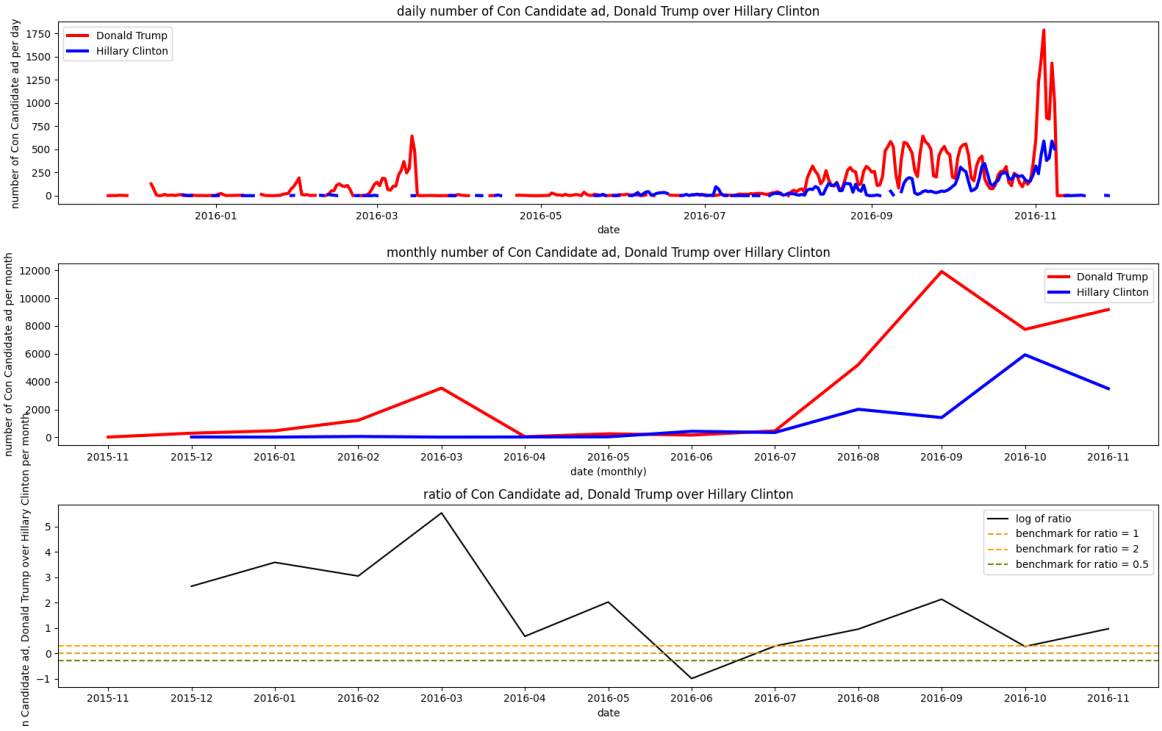
In terms of the number of ad on each candidate between Donald Trump and Hillary Clinton, we can see from the first graph below, that the total number of ad about Donald Trump doubles that of Hillary Clinton since the official party-endorsed presidential nominee announcements (July 2016), till the election day. During the period of primary debates, there were more TV ad on Hillary Clinton over Donald Trump initially, but they start to converge to each other as most of the major primary debates ended in March 2016. Here we use monthly total number of ad instead of daily, considering that there are days where no ad is on about at least one candidate and will cause less meaningful ratio.



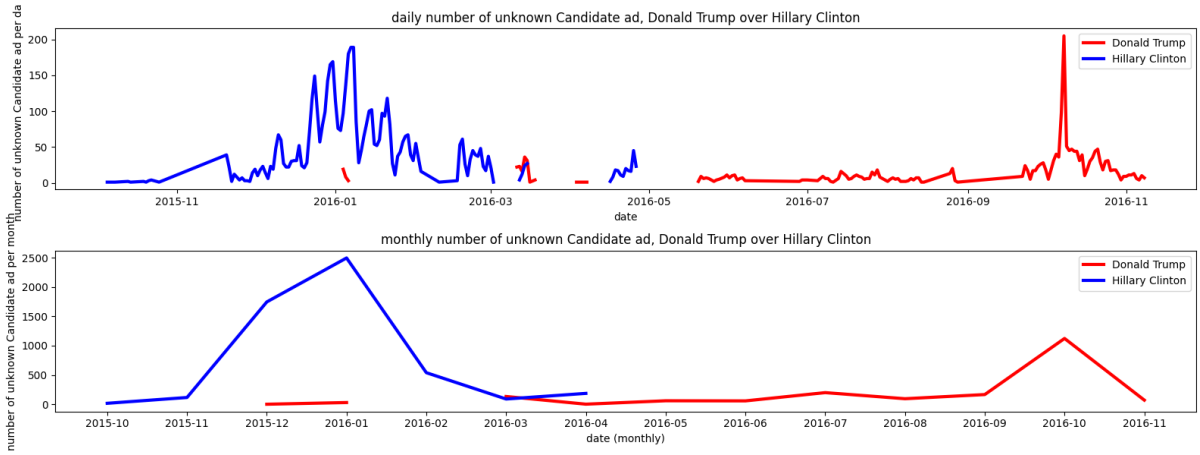
As we dive a bit deeper into the decomposition of ad by message type, we can see that from the graph below, that the number of ad conveying pro-candidate message (in support of the candidate) on Hillary Clinton, than the number on Donald Trump, both during primary debate season, and at the beginning of the presidential debate season. However, as the time gets closer to election, the number of ad containing pro-candidate message started to converge between these two candidates.



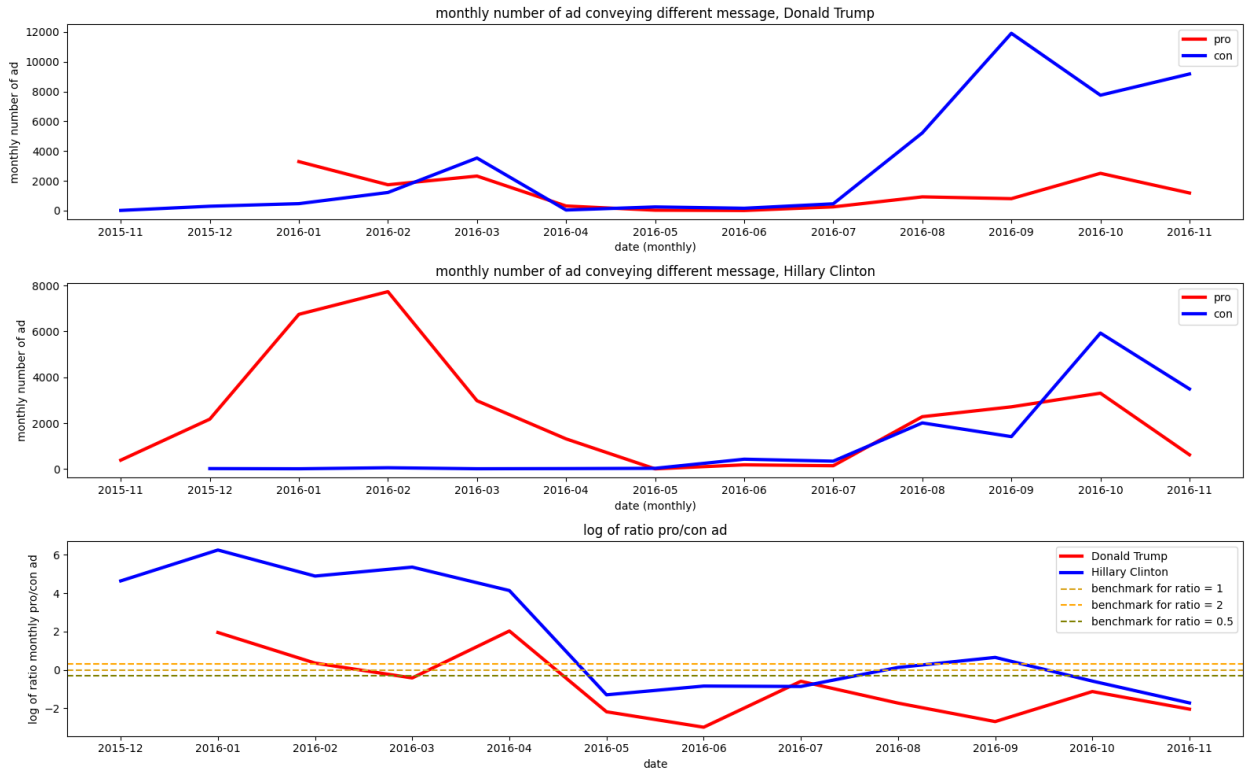
From the graph below, we can see that the number of ad conveying con-candidate message (against the candidate) on Donald Trump, is well above the number on Hillary Clinton, almost consistently through the whole 2016 presidential election cycle. It is especially the case during the primary debate season, the presidential debate season, and the period of time around final election.



The number of ad conveying message belongs to 'unknown' are relatively small, but from the graph we plot below. we notice one interesting thing we notice here is that the periods where ad conveying unknown type message happened for Clinton and Trump hardly overlap. Ad conveying unknown type message on Hilary occurred mostly during primary debate season, while for Trump those occurred the most during presidential debate season. Therefore, the ratio contains little information and is not plotted or analyzed.



According to the graph below, for Donald Trump, there has been more ad conveying message against him than in support of him during most of the primary debate season, the presidential debate season, and the period of time around final election. For Hillary Clinton herself, ad conveying message in support of her outweighed those against her during primary debates season, but got surpassed by those against her since the late end of presidential debates to the election day, potentially caused by more intense attack from her counterparty, and more intense discussion on her email scandal. There is no 'mix' type message for these two candidates, so no graph and discussion on that type of message.



Conclusions and Thoughts

Conclusion 1: We can conclude from the analysis that there is a trend in the number of TV ad distributed through the cycle of 2016 presidential election, where the number increases during primary debate season (roughly from December 2015 to March 2016), presidential debate season (August 2016 after the official announcement of presidential nominee by each party, to mid-November 2016 the end of last presidential debate), and the time period close to election day (mid-October 2016 to early November the election day). The number of TV ad dropped sharply after major primary debates ended. Candidates who still wanted to fight for the chance of being the party nominee may have a few more ad on TV (like Bernie Sanders), who announced drop-out of the presidential mostly will have no TV ad on presidential election after the announcement (like Marco Rubio).

Conclusion 2: There were more TV ad against Donald Trump, compared to against Hillary Clinton. And there were more TV ad against Donald Trump, than in support of Donald Trump. Both patterns showed consistency through the 2016 presidential election cycle, especially during the three periods mentioned in Conclusion 1 (primary debate season, presidential debate, and the time period close to election) when there were more political TV ad overall.

Conclusion 3: There were more TV ad airing in support of Hillary Clinton at first during primary debate season, but in the late cycle of the presidential election the number of TV ad against her became the majority.

Some causes that could be behind why there were more TV ad against Trump through out, or against Hillary in the late cycle of the 2016 presidential election, are 1) because they received more negative criticism on their performance, and 2) they received more attack from counterparty. Even though they sound similar, the logic behind these reasons can be completely different and will lead to different further conclusion. We may have a chance to explore this question, using column 'sponsers' and 'sponser_types', as these two columns can provide us more insights on the person/organization sitting behind those political TV ad that convey messages against some other candidates. It can be a really interesting follow-up project.

References

1. https://en.wikipedia.org/wiki/2016_Republican_Party_presidential_debates_and_forums
2. https://en.wikipedia.org/wiki/2016_Democratic_Party_presidential_debates_and_forums
3. https://en.wikipedia.org/wiki/2016_United_States_presidential_debates
4. https://www.npr.org/2016/03/15/470438649/marco-rubio-ends-his-presidential-campaign